STAR: Space mission Twin for Architecture Representation

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Space missions are becoming increasingly complex, largely due to the systemic nature of modern programs. Many of them include constellations with multi-instrument payloads, high on-board autonomy, and growing communication capacity. At the same time, these systems must be designed and delivered faster, demanding rapid iterations and data-driven operational decisions. This combination calls for advanced workflows that manage mission complexity by integrating design, verification, and operations support across the mission life-cycle.

This paper presents a Space mission Twin for Architecture Representation referred to as STAR. STAR is a digital twin framework, developed within Space It Up! (SIU) project, formulated to overcome the traditional design approach, supporting the space mission across all phases, including early design definition, solutions assessment, “what if?” scenarios validation, and operational life decision-making.

The tool uses a system-of-systems perspective, enabling analysis at multiple levels of granularity, from constellation design to subsystem behavior. The framework is modular and applicable to different mission scenarios, such as Earth Observation, In Orbit Services, and Space Exploration. STAR includes multiple interoperable modules to explore mission and subsystem domains. Its modular design allows each digital twin to be executed independently to investigate a specific driver or critical aspect, such as Electrical Power Subsystem (EPS) performance and the space environment interaction or combined with selected modules to study coupled effects across subsystems. When integrated as a whole, the framework supports a comprehensive mission assessment that connects design choices with operational implications within a single coherent framework. The tool provides capabilities ranging from attitude dynamics and sensor acquisition simulations to structural interactions with the environment, as well as deployable asset modelling, electric propulsion concepts, conjunction risk assessment and clock anomaly detection. Moreover, the modular architecture is inherently extensible, facilitating the future integration of additional modules to address critical and emerging aspects that are not yet covered by the current set. Finally, STAR aims to manage mission complexity by reducing risks and costs while improving performance from early design to operational life within a single environment.

The paper provides a technical presentation of STAR's modules and of the interfaces currently defined among them, showing how each module can be used either as a stand-alone capability or activated in connection with selected modules to support integrated mission analyses. The work is developed within Space It Up! (SIU), the Italian national initiative funded by the Italian Space Agency (ASI) and the Ministry of University and Research (MUR).